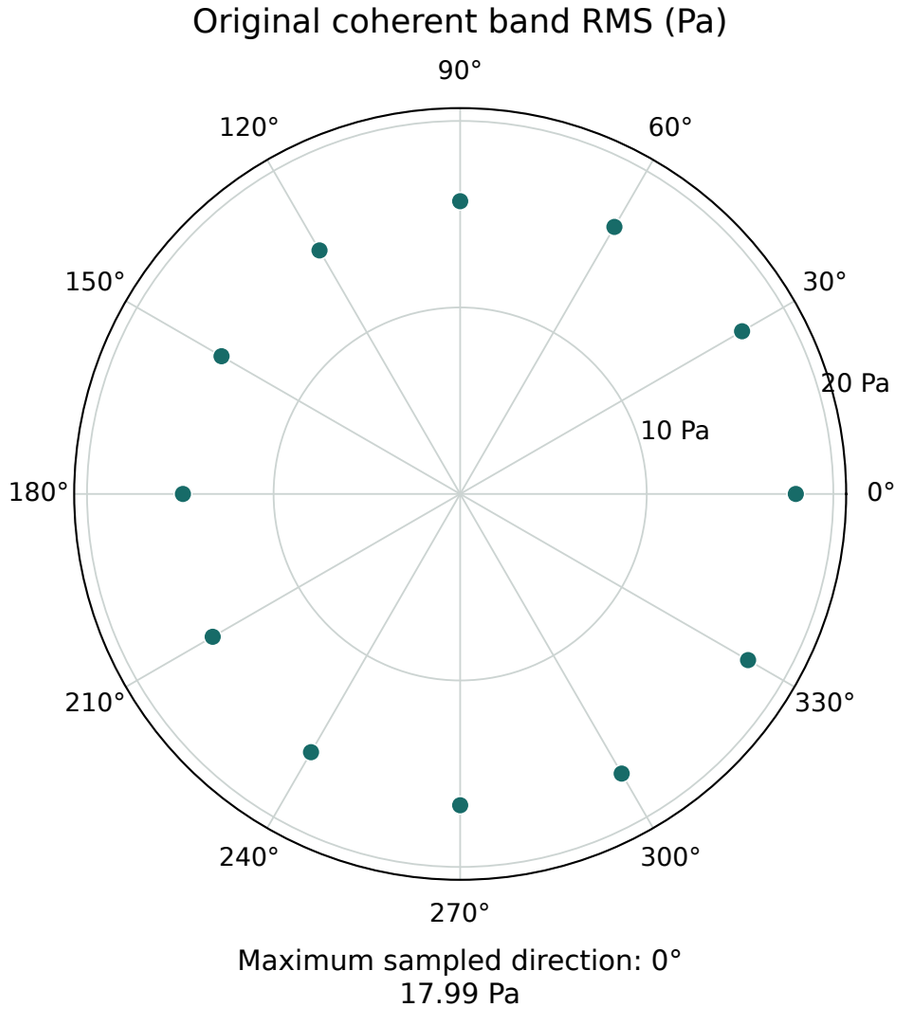
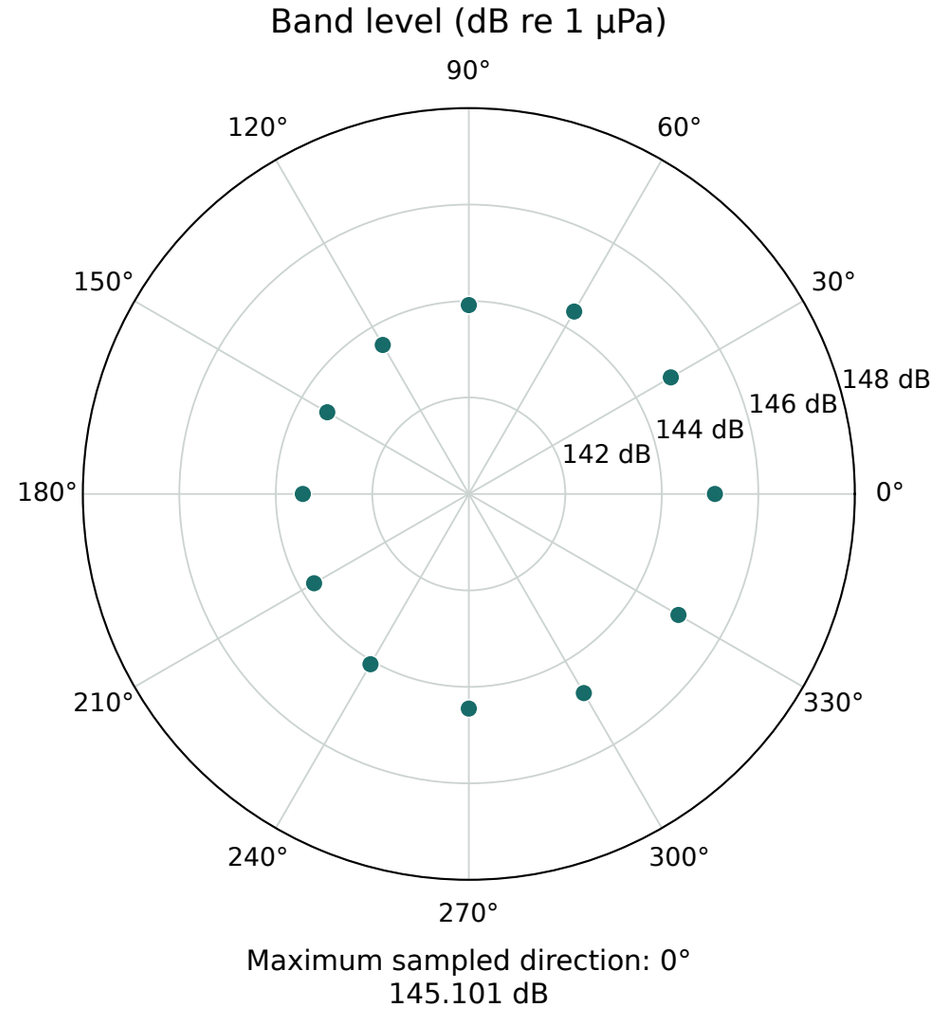
Selected-band directivity · 12 sampled observers

Case 4 · single-phase strict-2D WALE / stationary permeable contour

Samples 2.002–10 s · periodic Hann · retained band 0.625–25 Hz · receiver radius 1 m



Maximum sampled direction: 0°
17.99 Pa



Maximum sampled direction: 0°
145.101 dB

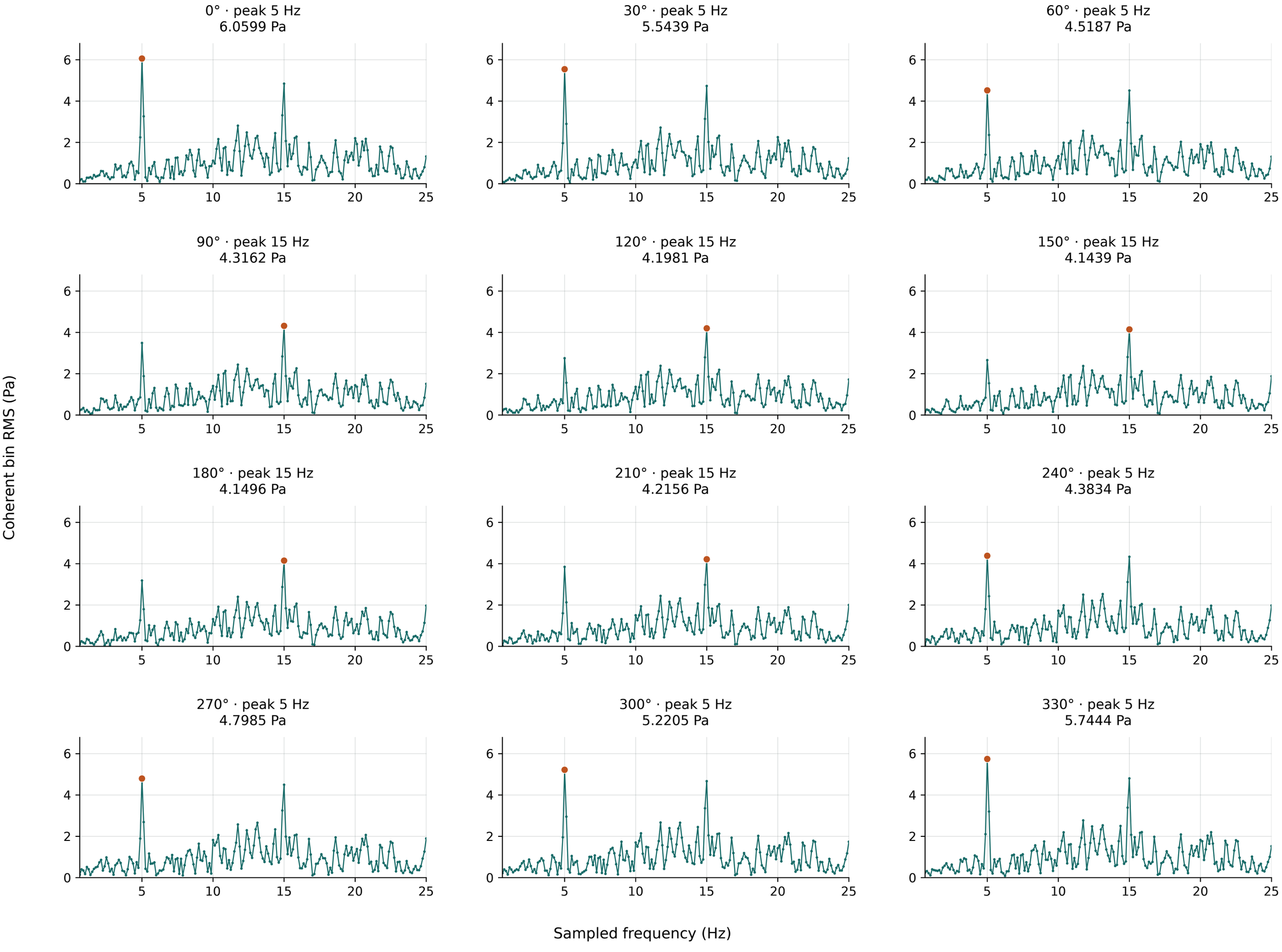
Unconnected markers: no angular interpolation. dB = 20 $\log_{10}$(original band RMS / 1 µPa).

Finite window-conditioned selected-band result; not a stationary-noise or absolute finite-turbine validation.

Coherent bin RMS · all 12 observers

Case 4 · single-phase strict-2D WALE / stationary permeable contour

Samples 2.002–10 s · periodic Hann · retained band 0.625–25 Hz · receiver radius 1 m



Bin RMS = √2 |c| (Pa); ENBW = 0.1875 Hz; bin spacing = 0.125 Hz. Orange marker: largest retained sampled bin.

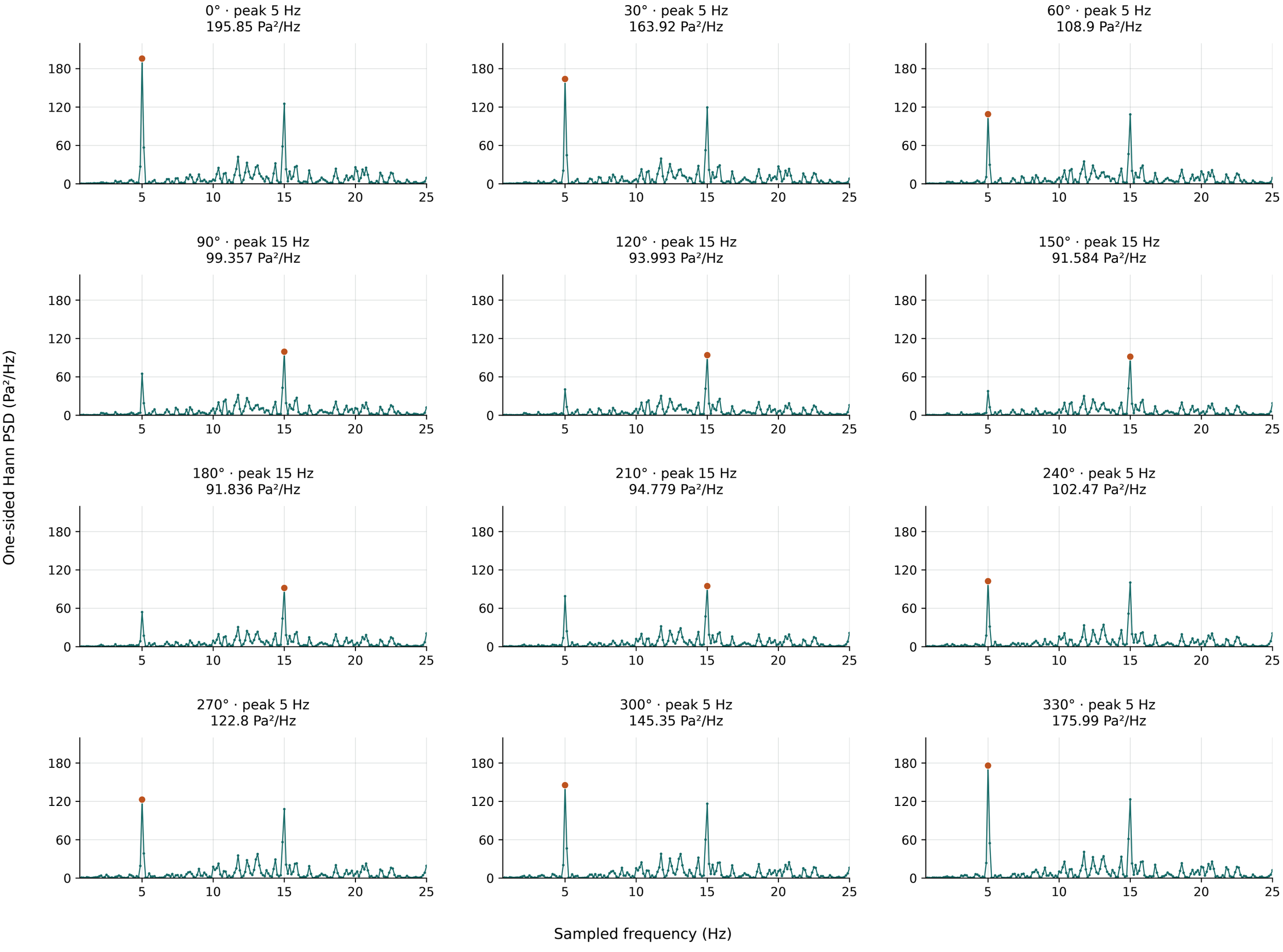
Single-record, propagated, window-conditioned estimate; no second Hann, Welch average or stationarity claim.

Selected-band PSD integral = coherent band mean square × df/ENBW (2/3 here). Peaks do not prove a blade-passing mechanism.

One-sided Hann PSD · all 12 observers

Case 4 · single-phase strict-2D WALE / stationary permeable contour

Samples 2.002–10 s · periodic Hann · retained band 0.625–25 Hz · receiver radius 1 m



PSD = 2|c|² / ENBW (Pa²/Hz); ENBW = 0.1875 Hz; bin spacing = 0.125 Hz. Orange marker: largest retained sampled bin.

Single-record, propagated, window-conditioned estimate; no second Hann, Welch average or stationarity claim.

Selected-band PSD integral = coherent band mean square × df/ENBW (2/3 here). Peaks do not prove a blade-passing mechanism.